UCCN 2003
UCCN 2243

TCP/IP Internetworking, Internetwork Principles & Practices

(Lecture 04b)

IPv6 Fundamental 2: IPv6 Transitions, IPv6 Tunnel

IPv6 Transitions

IPv6 Transition (from IPv4)

- IPv6 transition mechanisms are technologies to facilitate the transitioning of the Internet from its IPv4 infrastructure to the next generation addressing system of IPv6
- The Internet will be transformed after IPv6 fully replaces IPv4.
  - However, IPv4 is in no danger of disappearing overnight.
  - Rather, it will coexist with and then gradually be replaced by IPv6.
- This change has already begun, particularly in Europe, Japan, and Asia Pacific.
- These areas have been exhausting their allotted IPv4 addresses, which makes IPv6 all the more attractive.
- In North America, the U.S. Department of Defense (DoD) mandated in October 1, 2003, that all new equipment purchased must be IPv6-capable.
  - The DoD have switched entirely to IPv6 equipment by 2008.

IPv6 Deployments

- IPv6 deployments will occur piecewise from the edge network (end user).
  - Core infrastructure will only be changed when significant customer usage demands it.
    - Meaning IPv4 will still in core network for a while.
  - Whenever possible, devices and applications should be capable of both IPv4 & IPv6, to minimize the delays and potential failures inherent in translation points.

IPv4-IPv6 Transition/Coexistence

- A wide range of techniques have been identified and implemented, basically falling into three categories:
  - (1) Dual-stack techniques,
    - to allow IPv4 and IPv6 to co-exist in the same devices and networks
  - (2) Tunneling techniques,
    - to avoid order dependencies when upgrading hosts, routers, or regions
  - (3) Translation techniques,
    - to allow IPv6-only devices to communicate with IPv4-only devices
- Expect all of these to be used, in combination

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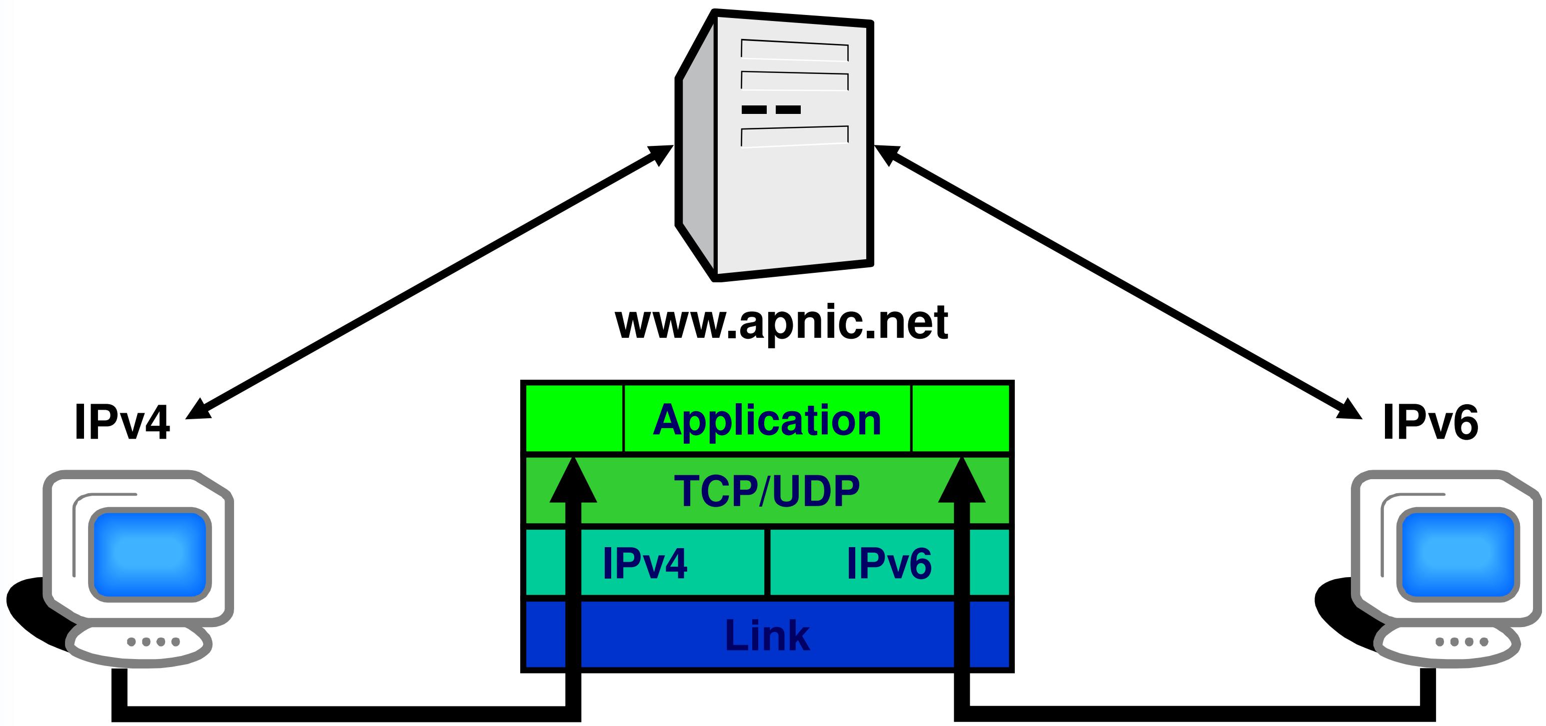
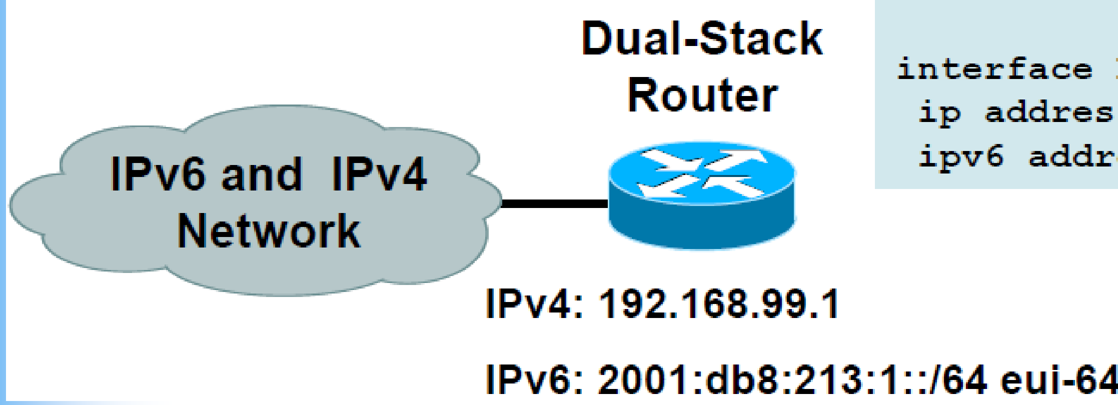
IPv6 transition – Dual Stack

- Dual stack hosts
  - Two TCP/IP stacks co-exists on one host
  - Supporting IPv4 and IPv6
  - Client uses whichever IP protocol it wishes



Cisco IOS Dual-Stack Configuration

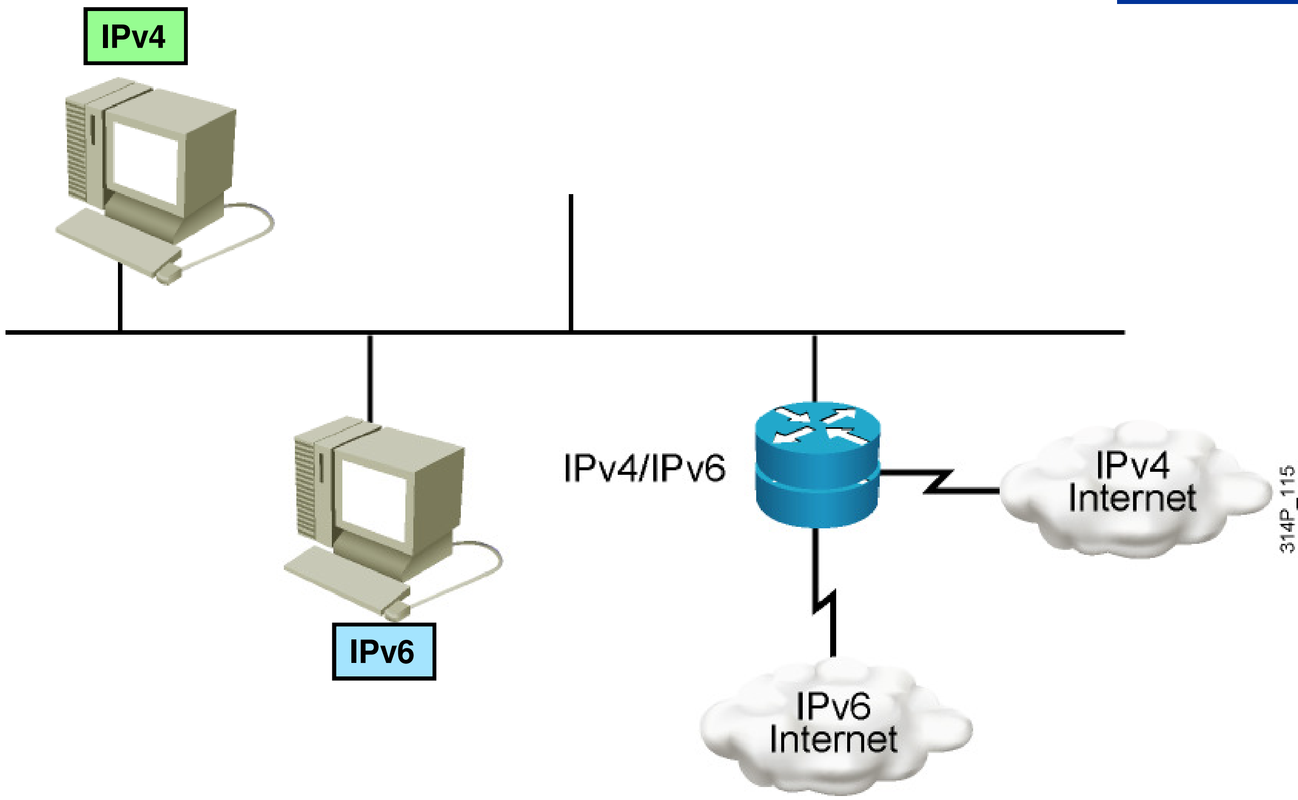



```
router#
ipv6 unicast-routing

interface Ethernet0
 ip address 192.168.99.1 255.255.255.0
 ipv6 address 2001:db8:213:1::/64 eui-64
```

- Cisco IOS Is IPv6-Enable:
  - If IPv4 and IPv6 are configured on one interface, the router is dual-stacked

Dual Stack Network



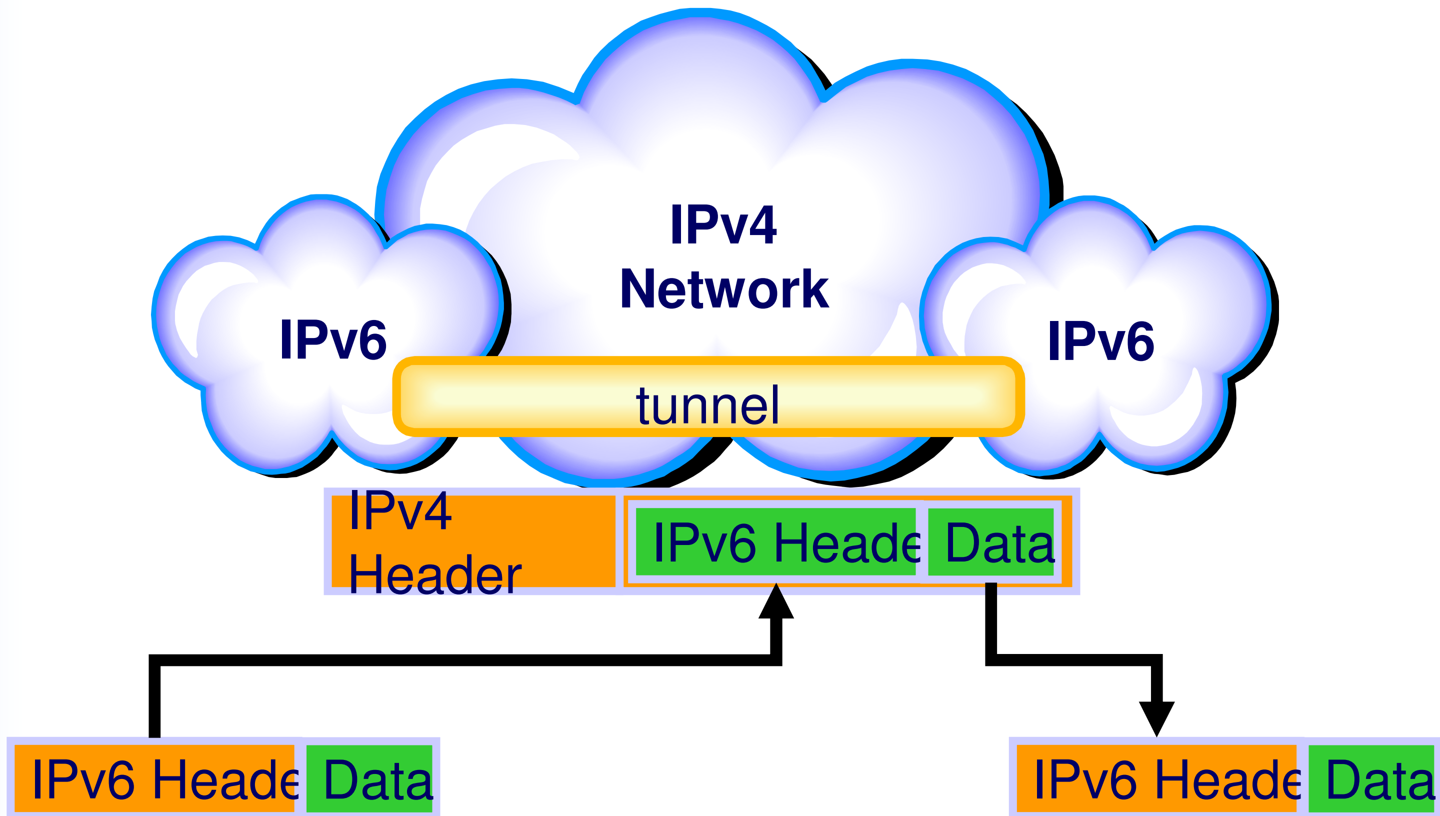
Dual stack is an integration method in which a node has implementation and connectivity to both an IPv4 and IPv6 network.

Comments on Dual Stack

- Features of the dual-stack method are as follows:
  - A dual-stack node chooses which stack to use based on the destination address.
  - A dual-stack node should prefer IPv6 when it is available.
  - IPv4-only applications continue to work as before.
  - This technique is well known and has been applied in the past for other protocol transitions. It enables gradual application upgrades, one by one, to IPv6.

IPv6 transition - Tunneling

- IPv6 tunnel over IPv4
  - IPv6 edge network going through IPv4 core network



IPv6 Transition - Translation

- This is a simple extension to NAT techniques, to translate header format as well as addresses
  - Sits between an IPv6 network and an IPv4 network.
  - The technique is to translate IPv6 packets into IPv4 packets and vice versa.
- It is not a popular mechanism.
  - This is really just a “temporary” solution for campus network.
  - “Going backward” (Isn’t that we are trying to escape from IPv4 NAT?)

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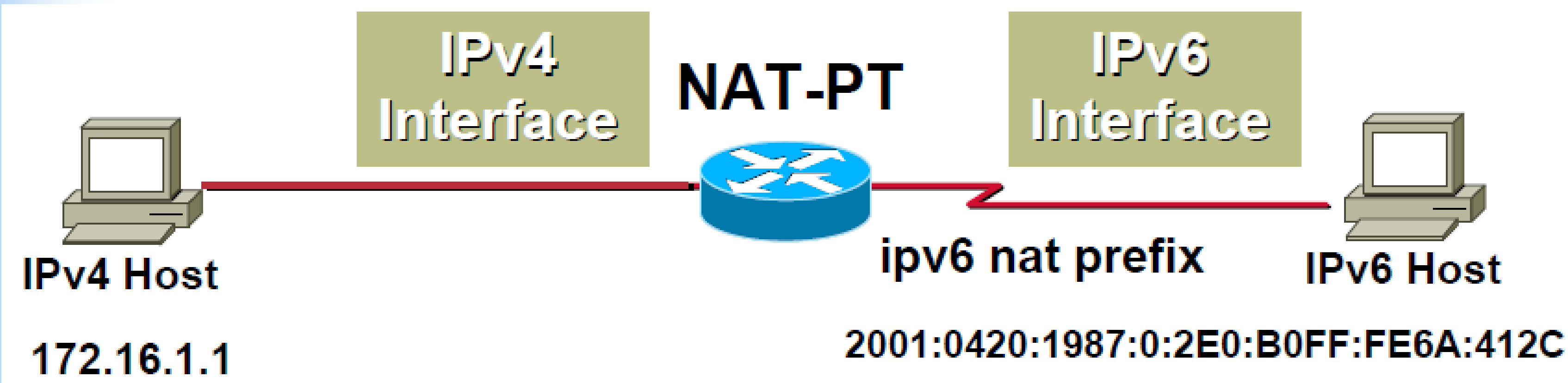
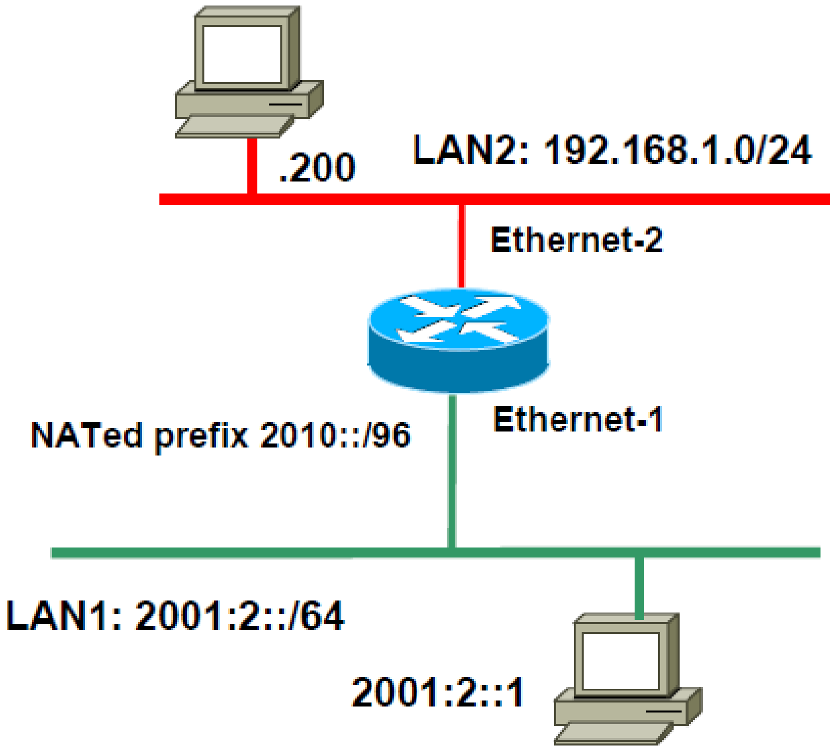
IPv6 Transition - Translation

- NAT-PT (Network Address Translation – Protocol Translation)
  - Allows native IPv6 hosts and applications to communicate with native IPv4 hosts and applications, and vice versa
  - Easy-to-use transition and co-existence solution



NAT-PT Configuration Example

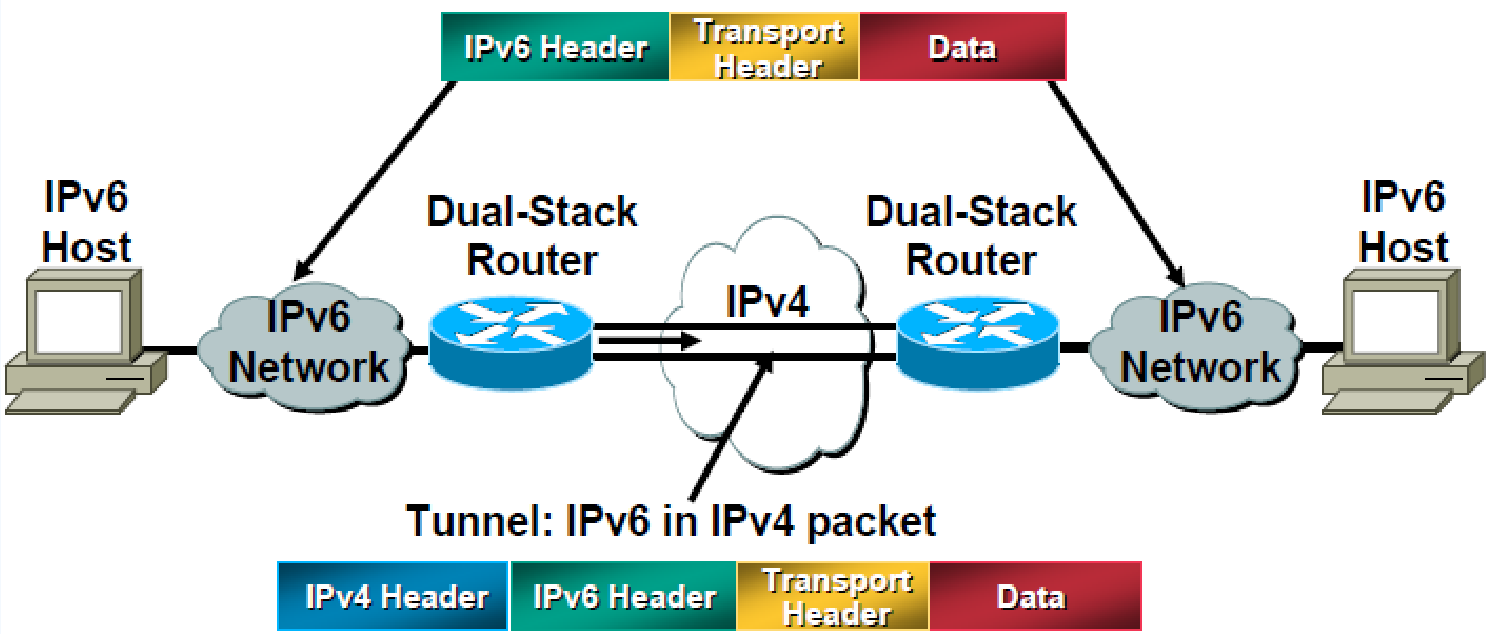



```
interface ethernet-1
   ipv6 address 2001:2::10/64
   ipv6 nat
!
interface ethernet-2
   ip address 192.168.1.1 255.255.255.0
   ipv6 nat prefix 2010::/96
   ipv6 nat
!
ipv6 nat v6v4 source 2001:2::1 192.168.2.1
ipv6 nat v4v6 source 192.168.1.200 2010::60
!
```

```
Router#show ipv6 nat trans
Prot  IPv4 source              IPv6 source
      IPv4 destination         IPv6 destination
---   ---                      ---
---   192.168.1.200            2010::60

---   192.168.2.1              2001:2::1
---   ---                      ---

Router#
```

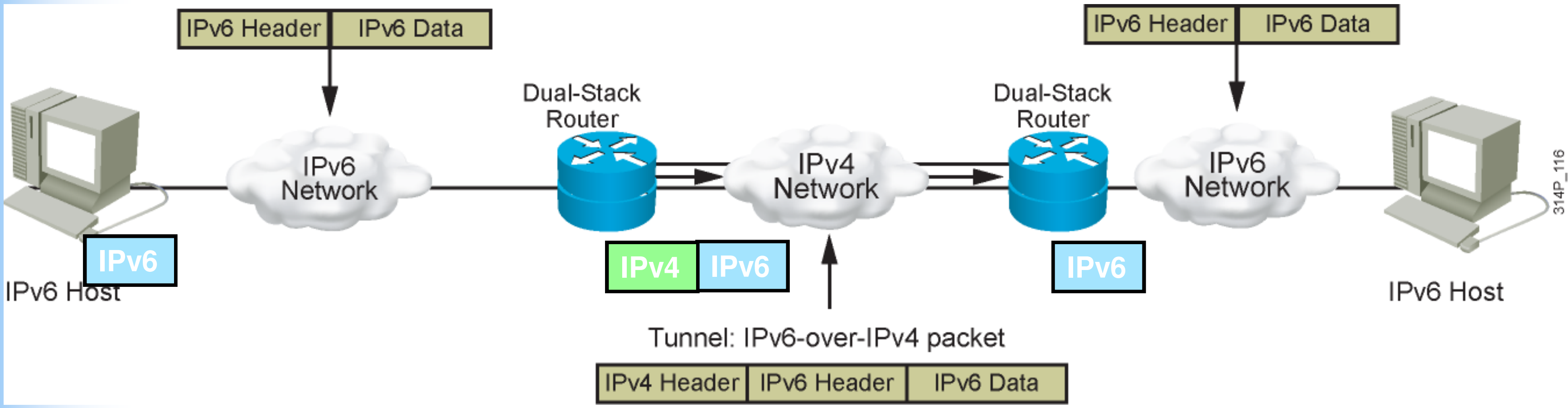
IPv6 Tunneling

IPv6 over IPv4 Tunnels



- Tunneling is encapsulating the IPv6 packet in the IPv4 packet.
- In the "tunnel", the packet has both IPv4 and IPv6 headers.
  - A 20-byte IPv4 header with no options and an IPv6 header and payload

Comments on IPv6 Tunneling

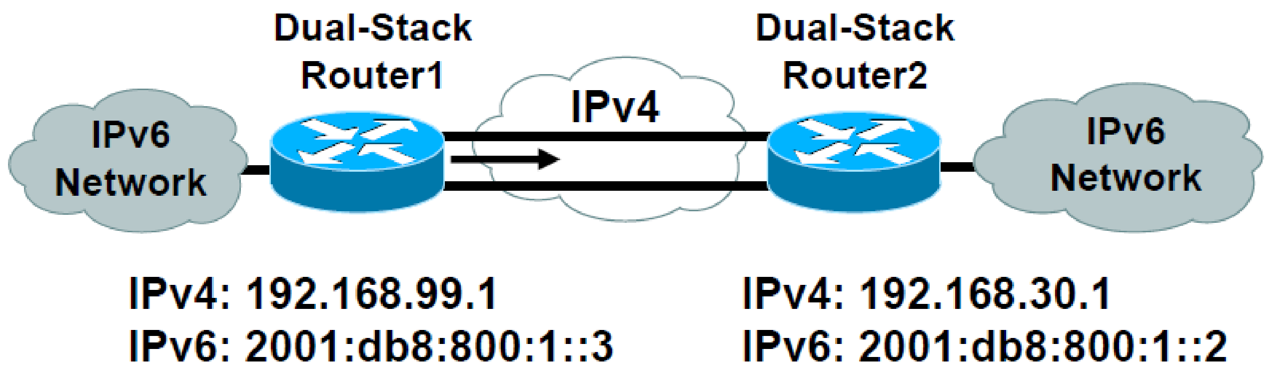


- Requires dual-stack routers
- This process connects IPv6 islands without needing to also convert an intermediary network to IPv6.
- Popular solution for IPv6 over WAN (IPv4 Internet).
- Tunneling is an intermediate integration and transition technique that should not be considered a final solution.
  - A native IPv6 architecture should be the ultimate goal.

Tunneling Techniques

- Many techniques are available to establish a tunnel:
- Manually configured:
  - Manual Tunnel (RFC 2893)
  - GRE (RFC 2473)
- Semi-automated
  - Tunnel broker
- Automatic
  - Compatible IPv4 (RFC 2893) : Deprecated
  - 6to4 (RFC 3056)
  - 6over4: Deprecated
  - ISATAP

Manually-Configured IPv6 over IPv4 Tunnel




```
router1#

interface Tunnel0
 ipv6 enable
 ipv6 address 2001:db8:c18:1::3/127
 tunnel source 192.168.99.1
 tunnel destination 192.168.30.1
 tunnel mode ipv6ip
```

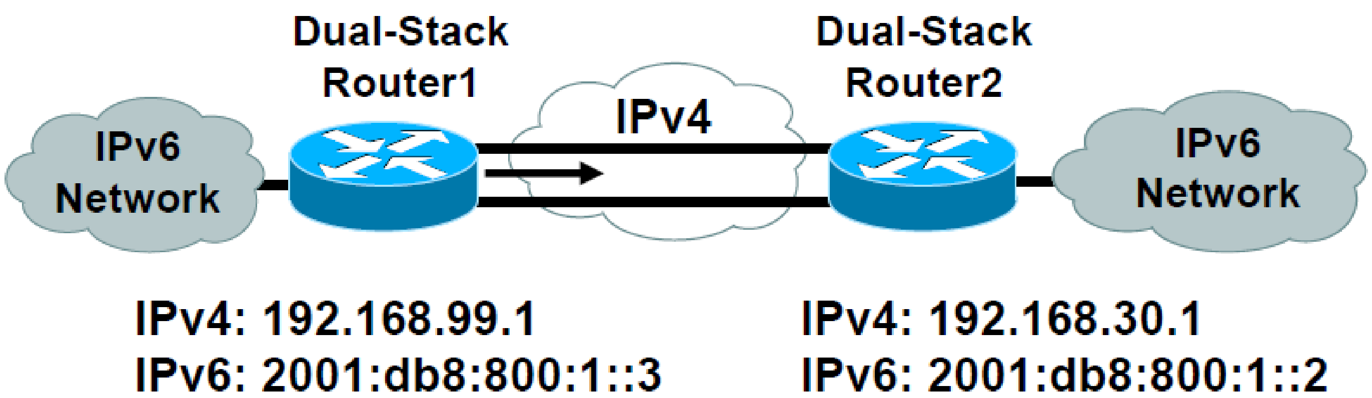
```
router2#

interface Tunnel0
 ipv6 enable
 ipv6 address 2001:db8:c18:1::2/127
 tunnel source 192.168.30.1
 tunnel destination 192.168.99.1
 tunnel mode ipv6ip
```

- Manually Configured tunnels require:
  - Dual stack end points
  - Both IPv4 and IPv6 addresses configured at each end.

IPv6 over IPv4 Tunnel Configuration

```
hostname R<No.>
!
ipv6 unicast-routing
!
interface Tunnel0
 no ip address
 ipv6 unnumbered Ethernet0/0
 tunnel source Ethernet1/0
 tunnel destination 192.168.30.110
 tunnel mode ipv6ip
!
interface Ethernet0/0
 no ip address
 ipv6 address 3FFE:B00:FFFF:1::/64 eui-64
 ipv6 address 3FFE:B00:FFFF:1001::/64 eui-64
 ipv6 address FEC0:0:0:1::/64 eui-64
 ipv6 enable
 ipv6 nd prefix 3FFE:B00:FFFF:1::/64 300 0
 ipv6 nd prefix 3FFE:B00:FFFF:1001::/64 300 300
 ipv6 nd prefix FEC0:0:0:1::/64 300 300
!
interface Ethernet1/0
 ip address 192.168.30.10<No.> 255.255.255.0
!
ipv6 route 3FFE:B00:FFFF:100A::/64 Tunnel0
```

Manually-Configured GRE Tunnel




```
router1#

interface Tunnel0
 ipv6 enable
 ipv6 address 2001:db8:c18:1::3/128
 tunnel source 192.168.99.1
 tunnel destination 192.168.30.1
 tunnel mode gre ipv6
```

```
router2#

interface Tunnel0
 ipv6 enable
 ipv6 address 2001:db8:c18:1::2/128
 tunnel source 192.168.30.1
 tunnel destination 192.168.99.1
 tunnel mode gre ipv6
```

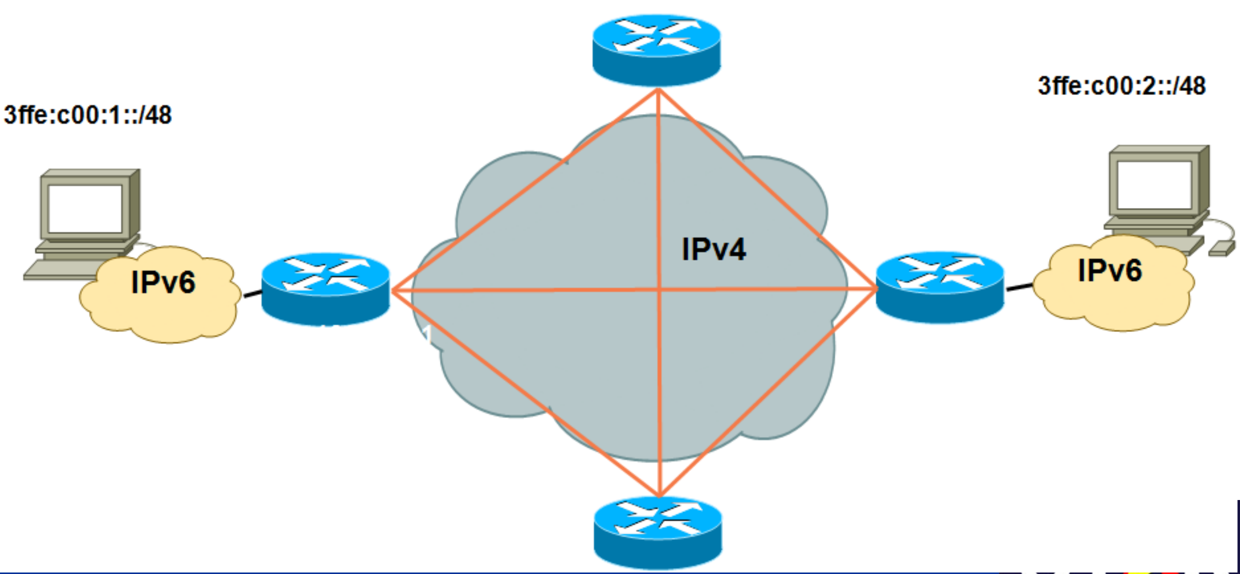
Comments on GRE Tunnel

- GRE = Generic Routing Encapsulation
- GRE is a tunneling protocol that was originally developed by Cisco, and it can do a few more things than IP-in-IP tunneling.
  - For example, you can also transport multicast traffic and IPv6 through a GRE tunnel.
- GRE header ≠ IP header

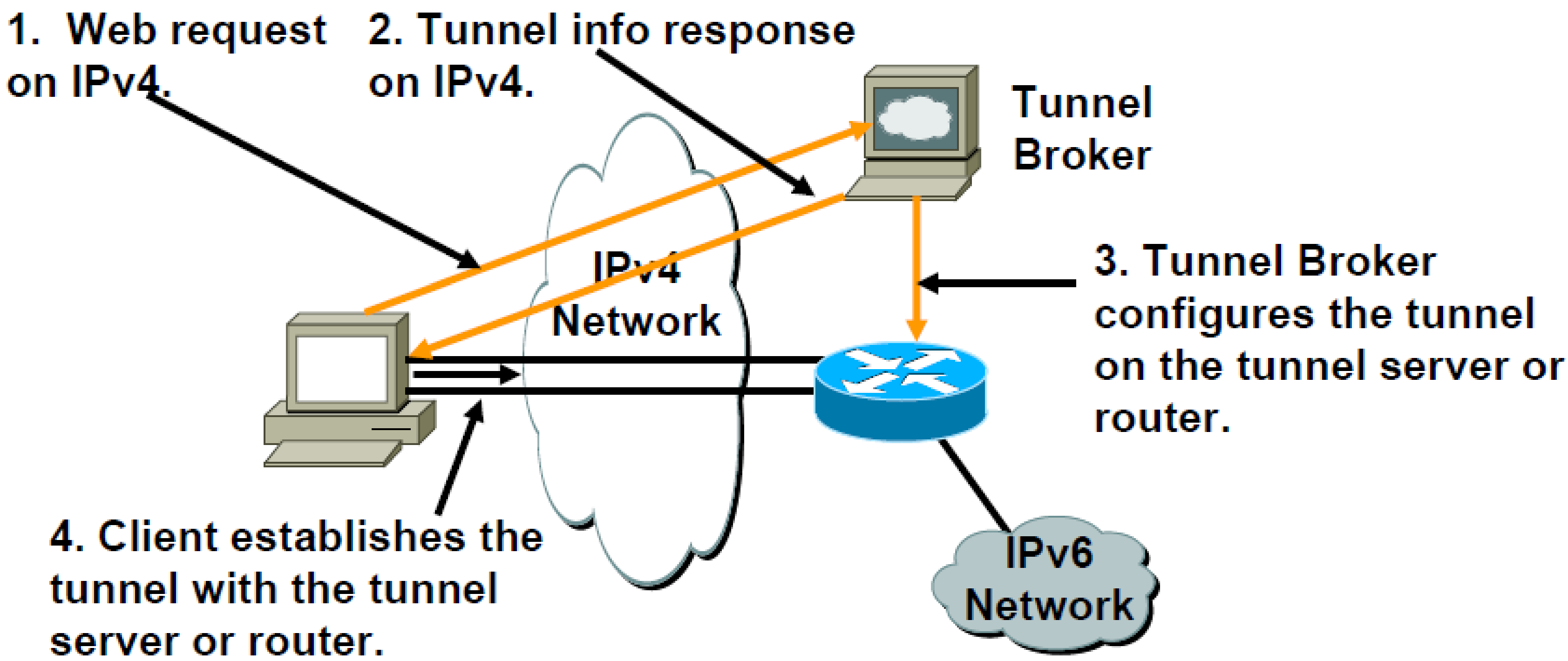
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
C	R	K	S	s	Recur			Flags					Version			Protocol															
Checksum																Offset															
Key																															
Sequence Number																															
Routing :::																															

Pros and Cons of Configured tunnels

Pros	Cons
As point to point links	Has to be configured and managed
Multicast	Inefficient traffic patterns
Real addresses	No keepalive mechanism, interface is always up



Tunnel Broker



- **Tunnel broker:**

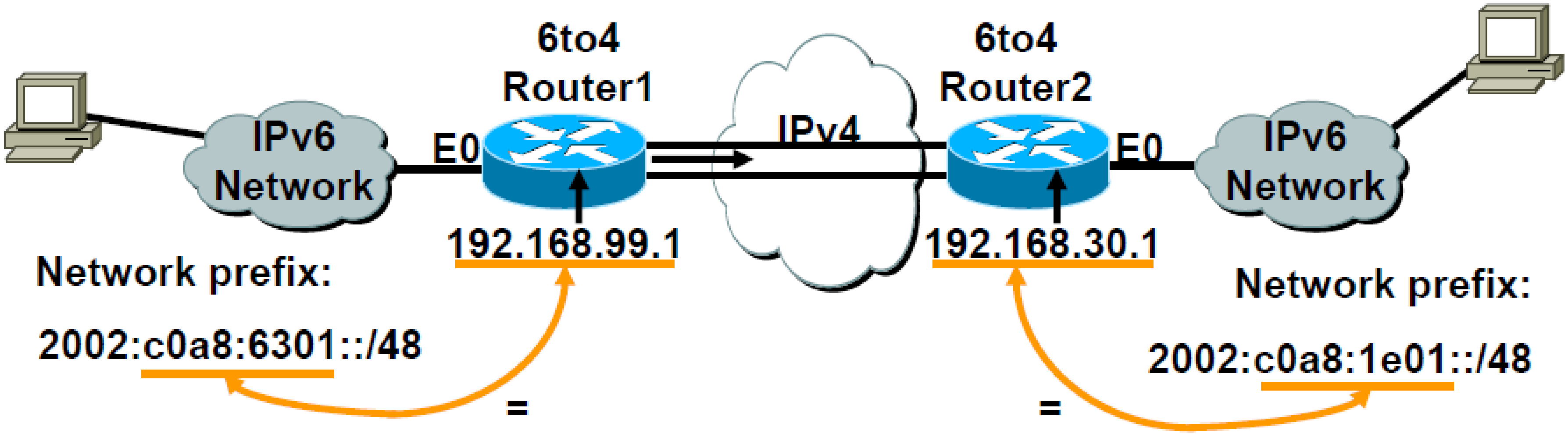
  Tunnel information is sent via http-ipv4

IPv6 6to4 Tunnel

- The key difference between automatic 6to4 tunnels and manually configured tunnels is that the tunnel is not point-to-point; it is point-to-multipoint.
- In automatic 6to4 tunnels, routers are not configured in pairs because they treat the IPv4 infrastructure as a virtual nonbroadcast multiaccess (NBMA) link.
- The IPv4 address embedded in the IPv6 address is used to find the other end of the automatic tunnel.
- The tunnel destination is determined by the IPv4 address of the border router extracted from the IPv6 address that starts with the prefix 2002::/16, where the format is 2002:*border-router-IPv4-address* ::/48.

IPv6 6to4 Tunnel

192.168.99.1 => C0.A8.63.01 (hex) 192.168.30.1 => C0.A8.1E.01 (hex)



- **6to4 Tunnel:**
  - Is an automatic tunnel method
  - Gives a prefix to the attached IPv6 network
  - 2002::/16 assigned to 6to4
  - Requires one global IPv4 address on each Ingress/Egress site

```
router2#
interface Loopback0
 ip address 192.168.30.1 255.255.255.0
 ipv6 address 2002:c0a8:1e01:1::/64 eui-64
interface Tunnel0
 no ip address
 ipv6 unnumbered Ethernet0
 tunnel source Loopback0
 tunnel mode ipv6ip 6to4

ipv6 route 2002::/16 Tunnel0
```

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IPv6 6to4 Tunnel

```
hostname R<No.>
!
ipv6 unicast-routing
!
interface Tunnel1
 no ip address
 no ip redirects
 ipv6 unnumbered Ethernet0/0
 tunnel source Ethernet1/0
 tunnel mode ipv6ip 6to4
!
interface Ethernet0/0
 no ip address
 ipv6 address 2002:<Hex Group No.>::/64 eui-64
 ipv6 enable
 ipv6 nd prefix 2002:<Hex Group No.>::/64 300 300
!
interface Ethernet1/0
 ip address 192.168.30.10<No.> 255.255.255.0
!
ipv6 route 2002::/16 Tunnel1
```

ISATAP – Intra Site Automatic Tunnel Addressing Protocol

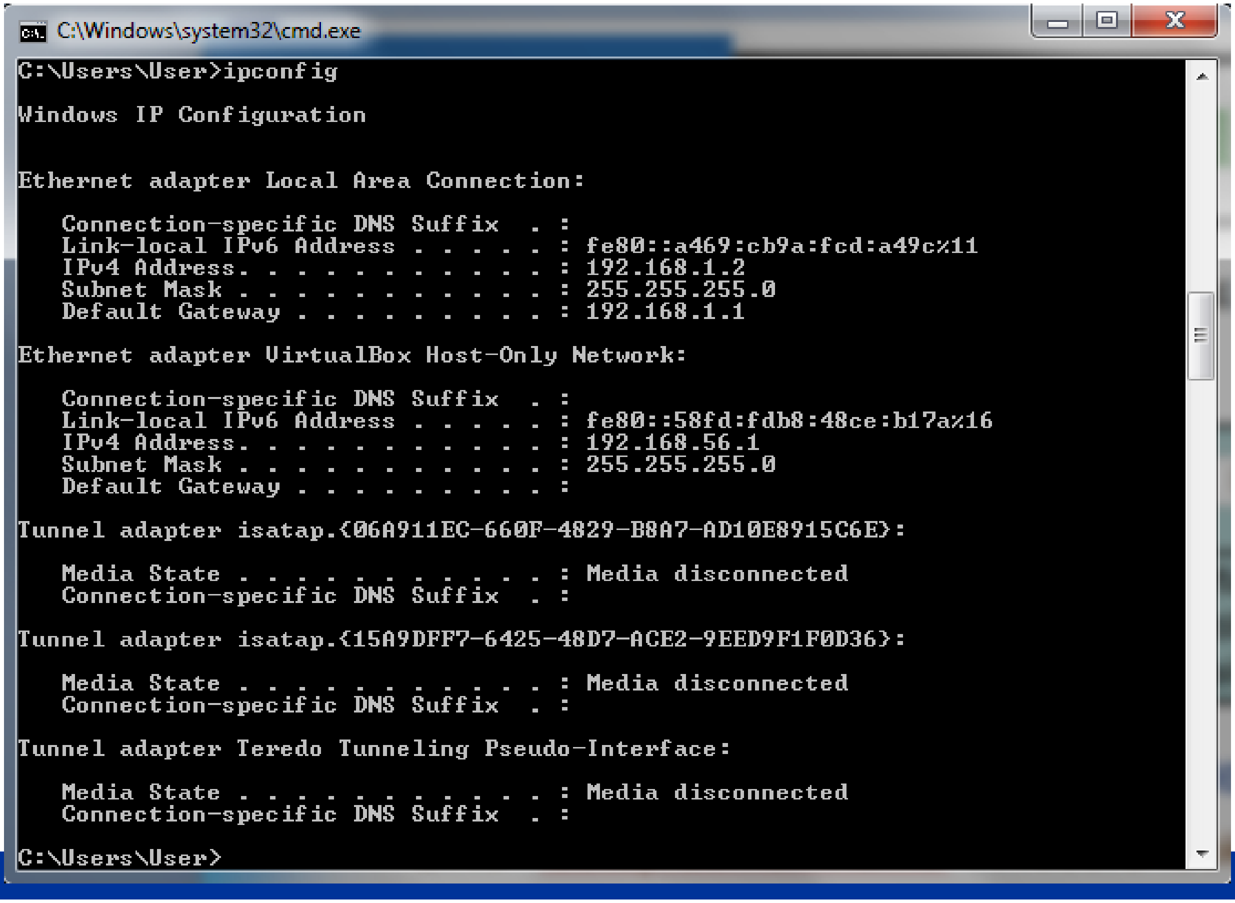
- Tunnelling of IPv6 in IPv4
- Single Administrative Domain
- Creates a virtual IPv6 link over the full IPv4 Network
- Automatic tunnelling is done by a specially formatted ISATAP address which includes:
  - A special ISATAP identifier
  - The IPv4 address of the node
- ISATAP nodes are dual stack

6to4 and ISATAP tunnels

- The main difference between automatic 6to4 tunnels and ISATAP tunnels is that the automatic 6to4 is **Inter-site** tunnel that allows IPv6 traffic between different sites where as ISATAP as the name specifies is for **Intra-site** which can be used for transporting IPv6 packets within a site, but not between sites.
- Another aspect is the address prefix used in sites, 6to4 sites uses addresses from 2002::/16 prefix where as ISATAP tunneling sites can use any IPv6 unicast address.

ISATAP is in your PC

- If you have installed Windows 7, you will see the isatap tunnel at the ipconfig output.



ISATAP Details

Use IANA's OUI 00-00-5E and encode IPv4 address as part of EUI-64

64-bit Unicast Prefix	0000:5EFE:	IPv4 Address
	32-bit	32-bit

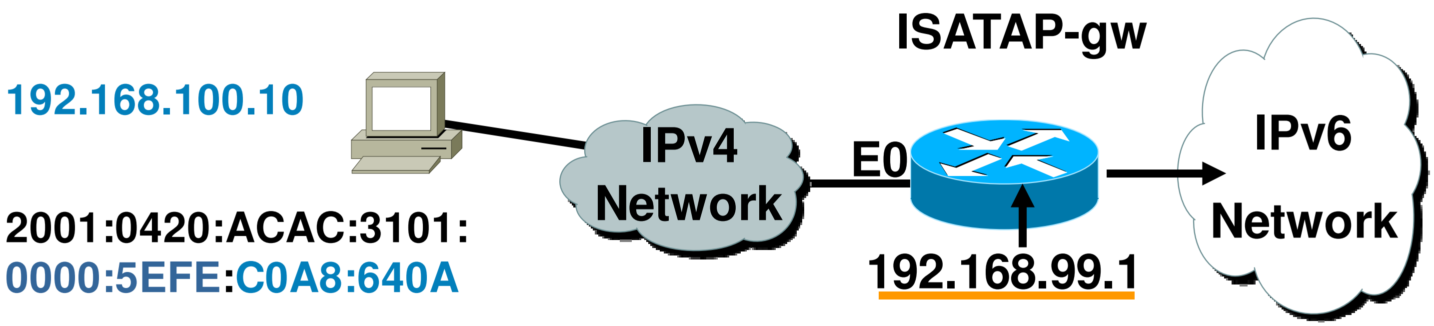
Interface Identifier (64 bits)

- Automatic discovery of ISATAP routers

Config isatap tunnel

```
C:\>netsh
netsh>int
netsh interface>ipv6
netsh interface>ipv6>install
netsh interface ipv6>isatap
netsh interface ipv6 isatap>set router isatap.sjtu.edu.cn enable
C:>ping6 ntp.buptnet.edu.cn

Pinging ntp.buptnet.edu.cn [2001:da8:202:10::2]
from 2001:da8:8000:d010:0:5efe:203.96.71.86 with 32 bytes of data:

Reply from 2001:da8:202:10::2: bytes=32 time=403ms
Reply from 2001:da8:202:10::2: bytes=32 time=407ms
Reply from 2001:da8:202:10::2: bytes=32 time=404ms
Reply from 2001:da8:202:10::2: bytes=32 time=406ms

Ping statistics for 2001:da8:202:10::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 403ms, Maximum = 407ms, Average = 405ms

C:\>
```

ISATAP Router




```
ISATAP-gw#
!
interface Ethernet0
 ip address 192.168.99.1 255.255.255.0
!
interface Tunnel0
 ipv6 address 2001:0420:ACAC:3101::/64 eui-64
 no ipv6 nd suppress-ra
 tunnel source Ethernet0
 tunnel mode ipv6ip isatap
```

Supported in Windows XP Pro SP1

The tunnel source command must point to an interface with an IPv4 address configured.

Configure the ISATAP IPv6 address, and prefixes to be advertised just as you would with a native IPv6 interface.

The IPv6 address has to be configured as an EUI-64 address since the last 32 bits in the interface identifier is used as the IPv4 destination address.